DO NOW — RECALL

Lesson 103 · Input and Output: Selecting Devices for Tas

Name: _____ Class: _____ Date: _____

1. What does an input device do?

2. What does an output device do?

3. What is an actuator?

4. A 2×3 display grid: how many cells?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Captures user/environment data.

2. Model answer

Communicates information or causes action.

3. Model answer

Output device controlling physical movement/action.

4. Model answer

Six.

By the end, you can…

• An input device captures data or a user action; an output device communicates information or changes physical conditions.

• A temperature sensor measures physical conditions; an actuator controls a physical action; a display shows values to the user.

• Choose devices by required accuracy, accessibility, environment and response time. A touchscreen can perform both input and output functions.

Vocabulary — know these terms

input — Data entering computer system.

output — Information/effect produced.

sensor — Measures a physical quantity.

actuator — Creates a physical action.

NEW LEARNING

Read, annotate and explain

Understand

An input device captures data or a user action; an output device communicates information or changes physical conditions.

Apply

A temperature sensor measures physical conditions; an actuator controls a physical action; a display shows values to the user.

Evaluate

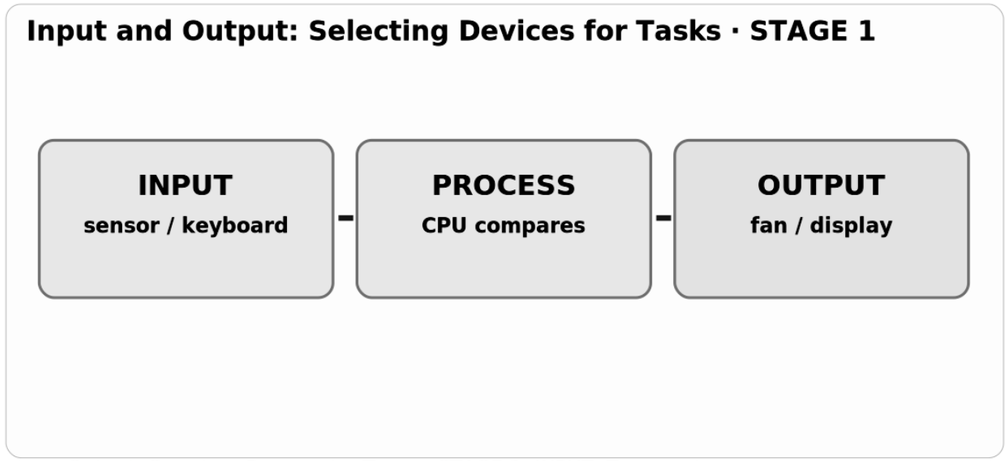
Choose devices by required accuracy, accessibility, environment and response time. A touchscreen can perform both input and output functions.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

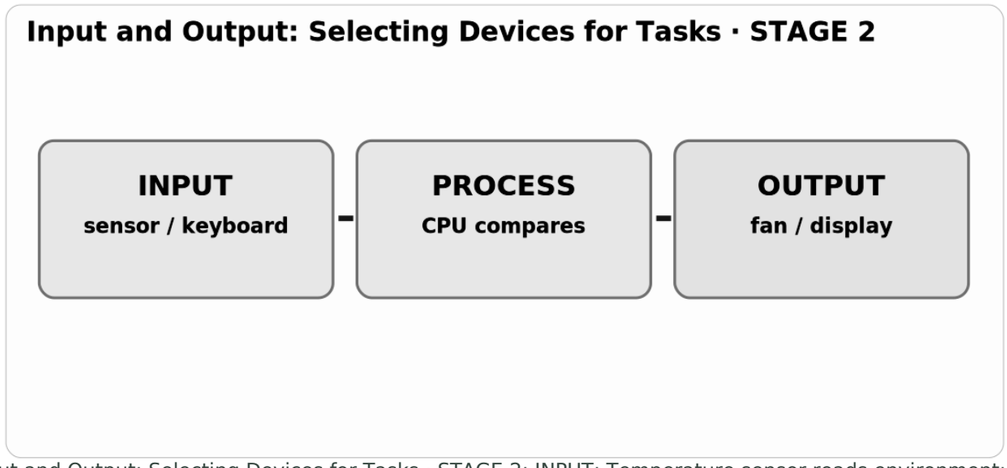
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Input and Output: Selecting Devices for Tasks · STAGE 1; INPUT: Temperature sensor reads environment

New Learning



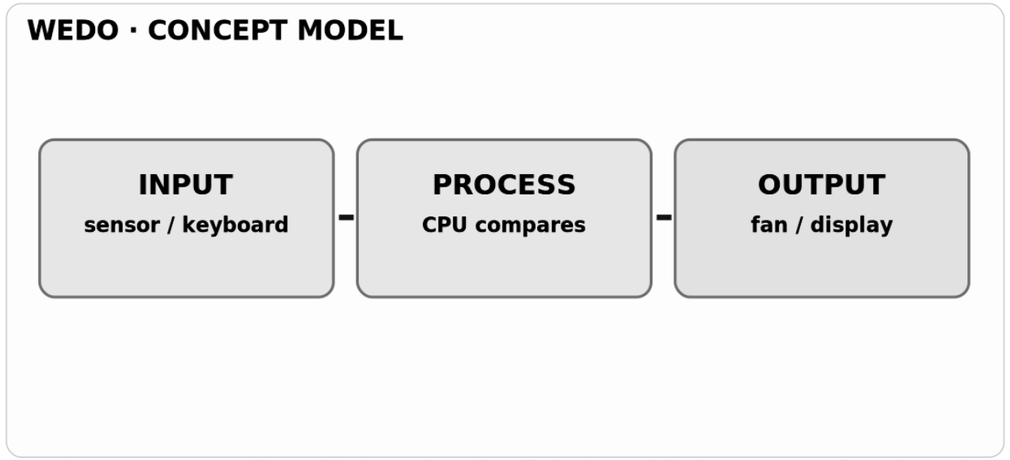
Input and Output: Selecting Devices for Tasks · STAGE 2; INPUT: Temperature sensor reads environment; PROCESSING: CPU compares 24°C with set point 20°C

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A library kiosk needs to identify books and display return confirmation. Select suitable devices.

Input barcode reader or RFID reader; output display and optional printer.



WEDO · CONCEPT MODEL; A library kiosk needs to identify books and display return confirmation. Select suitable devices.: Input barcode reader or RFID reader; output display and optional printer.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

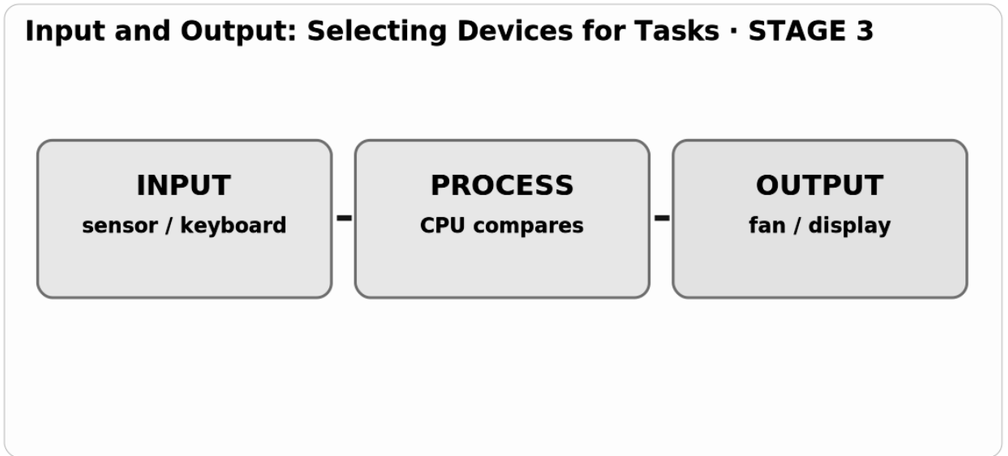
Guided 1 — CHECK

Explain the choice
Input barcode reader or RFID reader; output display and optional printer.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Input and Output: Selecting Devices for Tasks · STAGE 3; INPUT: Temperature sensor reads environment; PROCESSING: CPU compares 24°C with set point 20°C; OUTPUT: Fan actuator ON; display reports value

I DO — WORKED EXAMPLE

Follow the worked method

Recommend input and output for automated greenhouse temperature control.

Explain the choice

Input thermistor/temperature sensor; output fan/heater actuator, possibly a display.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

A touchscreen can only be an output device.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

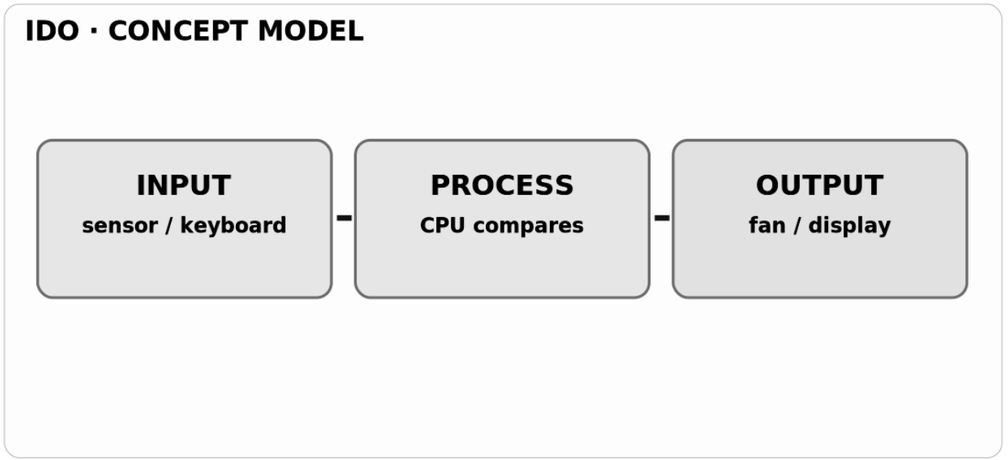
Design accessible touchscreen ticket kiosk: give two input and two output features and justify. [8]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Recommend input and output for automated greenhouse temperature control.

Input thermistor/temperature sensor; output fan/heater actuator, possibly a display.



IDO · CONCEPT MODEL; Recommend input and output for automated greenhouse temperature control.: Input thermistor/temperature sensor; output fan/heater actuator, possibly a display.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

A library kiosk needs to identify books and display return confirmation. Select suitable devices.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

For a room-temperature controller, choose appropriate input and output devices and explain roles. [6]

Question 2

Compare barcode reader and RFID for counting library books, linking benefits to context. [6]